$\therefore T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = T(x_1e_1 + x_2e_2) = T(x_1e_1) + T(x_2e_2)$ (From Condition 1)

$= x_1T(e_1) + x_2T(e_2)$ (From Condition 2) $= x_1y_1 + x_2y_2 = x_1\begin{bmatrix} 2 \\ 5 \end{bmatrix} + x_2\begin{bmatrix} -1 \\ 6 \end{bmatrix}$

$= \begin{bmatrix} 2x_1 \\ 5x_1 \end{bmatrix} + \begin{bmatrix} -x_2 \\ 6x_2 \end{bmatrix}$ (Scalar multiplication)

$= \begin{bmatrix} 2x_1 - x_2 \\ 5x_1 + 6x_2 \end{bmatrix}$ (Vector addition) □

20. Let $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $v_1 = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$, and $v_2 = \begin{bmatrix} 7 \\ -3 \end{bmatrix}$, and let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation that maps x into $x_1v_1 + x_2v_2$. Find a matrix A such that $T(x)$ is Ax for each x.

$\boxed{\text{Solution:}}$ $T(x) = x_1v_1 + x_2v_2 = x_1\begin{bmatrix} -2 \\ 5 \end{bmatrix} + x_2\begin{bmatrix} 7 \\ -3 \end{bmatrix} = \begin{bmatrix} -2 & 7 \\ 5 & -3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ (By defn. of matrix eqn. $Ax = b$)

$= \begin{bmatrix} -2 & 7 \\ 5 & -3 \end{bmatrix} x$ $\boxed{\therefore A = \begin{bmatrix} -2 & 7 \\ 5 & -3 \end{bmatrix}}$ □

In Exercises 21 and 22, mark each statement True or False. Justify each answer.

21) a. A linear transformation is a special type of function.

$\boxed{\text{Solution:}}$ On Page 66 in the grey box, it is clear that if any transformation follows the conditions (i) and (ii), then it is linear. On Page 64, it is mentioned that a transformation (or function or mapping) T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a rule that assigns to each vector x in $\mathbb{R}^n$ a vector $T(x)$ in $\mathbb{R}^m$.

$\boxed{\therefore \text{A linear transformation is a special type of function}}$. □

The given statement is True.

b) If A is a 3×5 matrix and T is a transformation defined by $T(x) = Ax$, then the domain of T is $\mathbb{R}^3$.

$\boxed{\text{Solution:}}$ Under the Section "Matrix Transformations" on Page 64, it is mentioned that if T is a transformation and A is a $m\times n$ matrix, the domain of T is $\mathbb{R}^n$ when A has n columns and the codomain of T is $\mathbb{R}^m$ when each column of A has m entries. The range of T is the set of all linear combinations of the columns of A, because each image $T(x)$ is of the form Ax.

$\therefore$ The domain of T must be $\mathbb{R}^5$. $\boxed{\text{The given statement is False.}}$ □